CHAPTER 4

Derivation, convexity and continuity

S29, S31, S32; recaptures S33 and S40

Before you start · Diversion and Curve Reading

Without a calculator, in six minutes. Respond briefly. Mobilize the derivatives of first and the signboards. Then choose the targeted review indicated under the questions.

1. Derivating x^3.

2. Derivating $(x+1)e^x$.

3. Give the sign of $(x-2)(x+3)$ on $]-3;2[$.

4. What is represented by $f'(a)$ Graphically?

5. Give an equation of tangent in a.

6. Give a non-derivable function in 0.

Responses

$3x^2$; $(x+2)e^x$; negative ; guiding coefficient of the tangent ; $y = f(a) + f'(a)(x-a)$; $|x|$.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Use the rule Power diversion. **2.** Write separately the two terms of the derivative of the product; R35. **3.** Make a signboard of both factors. **4.** Calculate a slope with two points; R37. **5.** Use the value and slope at the contact point; R37. **6.** Compare the slopes to the left and to the right of absolute value. R-questions serve as support for resuming with the teacher, even if their collective ritual is planned later.

Objectives

Derivating a compound; studying variations and convexity; demonstrating existence and uniqueness; approaching a solution.

Course notes

Deriving a compound

If u is derivable on I and v Derivable over an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. The fields must be checked prior to calculation.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad \left(\frac{1}{u}\right)' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is derivable for $x > -\frac{1}{3}$ and $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of f' determines changes in f; the sign of f do not determine them.

Convexity

A function f is convex over an interval I if his curve is under his ropes: $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a two-fold detritable function, equivalences are allowed: f convex; f' increasing; $f'' \geq 0$; the curve is above all its tangents. Inequalities are reversed for a function Concave. A point of inflection is a point where the curve changes convexity.

Demonstration: position relative to tangents. Suppose $f'' \geq 0$. Fixes a and positons $g(x) = f(x) - f(a) - f'(a)(x-a)$. Like f' is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and ≥ 0 for $x \geq a$. Thus g reached its minimum $g(a) = 0$ in a, therefore $f(x) \geq f(a) + f'(a)(x-a)$.

Examples. x^4 is convex and $f''(0) = 0$, but 0 is not an apse of inflection. x^3 exchange rate of convexity in 0. The convexity of the exponential and its tangent in 0 give $e^x \geq 1 + x$.

Continuity and equations

f is continuous in a if $\lim_{x \to a} f(x) = f(a)$. Any drifting function is continuous. Reciprocality is false: $|x|$ is continuous in 0 but not driftable in 0.

Theorem of intermediate values

If f is continuous on $[a,b]$, any real between $f(a)$ and $f(b)$ has at least one preimage in $[a,b]$. If f is also strictly monotonous, this preimage is unique.
work on an open interval using boundary at bounds.

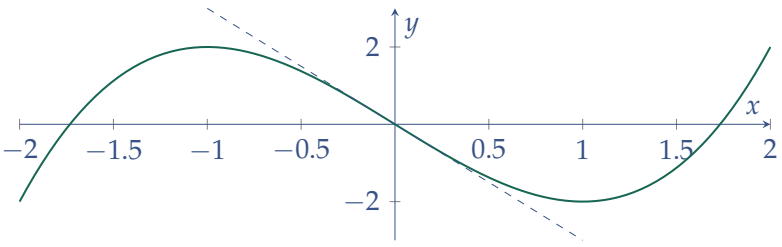
Prepared example. $f(x) = x^3 + x - 1$ is continuous on $[0,1]$. Like $f(0) = -1$ and $f(1) = 1$, the VIT gives a zero in $]0,1[$. Since $f'(x) = 3x^2 + 1 > 0$, it's unique. $f(0{,}68) < 0 < f(0{,}69)$ provides a framework, without replacing justifications of existence and uniqueness.

A function defined by pieces. For $x \neq 1$, pose $g(x) = \frac{x^2-1}{x-1}$, and pose $g(1) = a$. Like $g(x) = x + 1$ when $x \neq 1$, $\lim_{x \to 1} g(x) = 2$. The function is continuous in 1 if and only if $a = 2$. The point value must therefore be compared with the limit, not replaced by it without justification.

Continuations and continuity. If $u_n \to \ell$ and f is continuous in ℓ, then $f(u_n) \to f(\ell)$. For a recurring sequence convergent $u_{n+1} = f(u_n)$, we get $\ell = f(\ell)$.

Dichotomy

For f continues and $f(a)f(b) < 0$, calculate $m = \frac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep half of which the extremities have images of opposite signs. n steps, length is $\frac{b-}{2^n}$. The environment The method does not guarantee unity; it proves itself separately.



$y = x^3 - 3x$ and its tangent with inflection.

Linking calculation, curve and argument

4.1 The inside factor and the field

First use: S29.

Example Solved · A two-step composition

For $f(x) = \sqrt{1+x^2}$, the argument is always strictly positive: f is derivable on $\mathbb{R}$. The external function has for derivative $t \mapsto 1/(2\sqrt{t})$, the inner function is derivative 2 x. Their product gives $f'(x) = x/\sqrt{1+x^2}$. The fraction is negative for $x < 0$, positive for $x > 0$: f Decline and then grow.

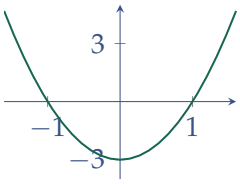
A1 · Immediate application Determine the differentiability domain and derivative of $g(x) = \sqrt{2x-1}$ and then $h(x) = e^{1-x^2}$. Check the sign of g'.

A2 · Repair a derivative "For $f(x) = (3x+1)^4$, we have $f'(x) = 4(3x+1)^3$. » Explain the factor and propose a verification by development or rate of increase in 0.

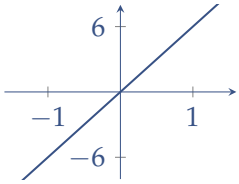
4.2 Associate f, f' and f''

First use: S29 and S31.

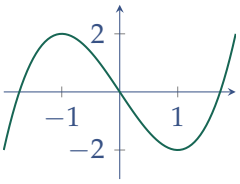
Curve A



Curve B



Curve C



Example resolved · A positive and decreasing function

For $g(x) = e^{-x}$, g is positive, $g' = -e^{-x} < 0$ and $g'' = e^{-x} > 0$. The curve goes down, but its negative slopes increase to 0: it is convex. The three informations are compatible.

A3 · Read before calculating The three curves correspond, in an order to find, to $f(x) = x^3 - 3x$, f' and f''. Associate them using zeros, extrema and changes in convexity; then control by bypass.

Property · Two readings not to be confused

$f'(a) > 0$ describes a positive slope of f, not a positive order. $f''(a) > 0$ indicates locally that the slopes increase; this does not mean that f increase. e^{-x} Decline and stay convex.

4.3 Between a tangent and a rope

First use: S31; take-back S33.

Example resolved · Two frameworks of x^2

On $[0;2]$, the rope joining $(0;0)$ to $(2;4)$ a for equation $y = 2x$; the tangent in 1 has for equation $y = 2x - 1$. Convexity gives 2 $x - 1 \leq x^2 \leq 2x$. Differences are established: $x^2 - (2x - 1) = (x - 1)^2 \geq 0$ and $2x - x^2 = x(2 - x) \geq 0$ over this interval.

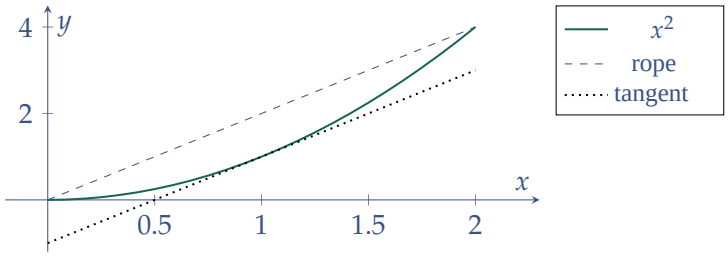


Figure 4.1. Compare the three ordered for $x = \frac{1}{2}$, then justify on all $[0;2]$.

A4 · Compare three functions For x^4, x^3 and $|x|$, study in 0 continuity, differentiability and change for both polynomials, explain why the cancellation of the second derivative does not
It is not enough to conclude.

4.4 Existence and uniqueness are two conclusions

First use: S32 and S33.

Example Solved · Write an Approached Resolution

Let $F(x) = x^3 + x - 1$. It is a polynomial, so it is continuous. $F(0) < 0 < F(1)$ gives a solution in $]0;1[$ by the VIT. $F'(x) = 3x^2 + 1 > 0$ on $\mathbb{R}$ ensures that there can't be two. Calculation $F(0{,}68) < 0 < F(0{,}69)$ locates this solution, but does not replace the existence or Oneness.

A5 · Two counter-examples The function is -1 for $x < 0$ and 1 for $x \geq$ Does it change the sign without cancelling? What hypothesis of the VIT lacks? For $x^2 = 1$, what hypothesis is missing for assert a single solution on $\mathbb{R}$?

4.5 Keep information at each stage

First use: S33 and S40.

Property · Invariant dichotomy

For a continuous function, maintain $F(a)F(b) \leq 0$ keeps at least one zero in $[a;b]$. À each step, the length is divided by 2. The algorithm gives a frame; uniqueness shall be established separately.

Example resolved · A first interval

For $F(x) = x^2 - 2$ on [1;2], $F(1) = -1$, $F(2) = 2$. In the middle $m = 3/2$, $F(m) = 1/4 > 0$; the half that keeps the change of sign is $[1;3/2]$. Its width is 1/2.

A6 · Choice of strategy Demonstrate that $e^{-x} = x$ has a single solution in $[0;1]$. Perform two dichotomy steps, then determine how many steps guarantee a range width at plus 1/128.

A first step: Bring the equation to the cancellation of a function. The number of steps comes from the width, not values of the function.

Wait a minute at the courthouse. Present the existence of a solution in one sentence containing the interval, the continuity and values at the ends; present unity in another sentence.

Review postponed · S47 and then S66 Apply the same arguments to a logarithmic function, then to an area dependent on a bound.

Research activity

Derivatives and graphical readings · 35 min.

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. We note $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$ and $f''(x)$, then draw up their signboards.
2. Deduct from the changes in f and its intervals of convexity and concavity.
3. Determine the coordinates of the deflection point of $\mathcal{C}_f$ and an equation of tangent at this point.
4. Using the limits, the extrema and the theorem of intermediate values, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure the uniqueness of a solution.

Progressive exercises

E1 · Direct application. We define on $\mathbb{R}$ functions $f(x) = (2x-1)^5$ and $g(x) = e^{-x^2}$. Calculate $f'(x)$ and $g'(x)$.

E2 · Application and justification. Let f the function defined on $\mathbb{R}$ by $f(x) = x^4 - 6x^2$. Determine the intervals over which f is convex or concave.

E3 · Method to be organised. Note $\mathcal{C}$ the exponential function curve. Determine an equation
1. With the help of convexity, deduct from it an inequality valid for all real x.

E4 · Choice of strategy. Demonstrate that the equation $x^3 + 2x = 5$ admits a single real solution. Frame this solution between two consecutive integers.

E5 · Synthesis or demonstration. A dichotomy algorithm starts from the interval $[0;2]$ and divides its length by two at each step. Determine the minimum number of steps ensuring an interval of a length not exceeding 10^{-3}.

Work in the manual or annal

Barbazo, p. 132, Exercise 29. In class after studying convexity: questions 1 and 2. At home: The aim is to distinguish the sign from the one in question 3 to 5. f', which governs variations, that of f'', which governs convexity.

Duty at home

DM4 · One maximum and one equation · 60 min.

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+1)e^{-x}$. Its representative curve is noted $\mathcal{C}_f$.

1. Calculate $f(0)$ and limit of f in $+\infty$. Calculate $f'(x)$, then draw up the table of variations of f.

2. Calculate $f''(x)$. Study the convexity of f and determine the coordinates of the possible point Inflection of $\mathcal{C}_f$.

3. Demonstrate that the equation $f(x) = \frac{1}{2}$ admits a single solution α on $[0;+\infty[$.

4. Write a Python function that uses dichotomy on $[0;2]$ and returns two reals a and b such that $a \leq \alpha \leq b$ and $b - a \leq 10^{-3}$. Justify the choice of the initial interval.

5. Determine an equation of tangent to $\mathcal{C}_f$ at abscisse point 0. The position of the curve by in relation to this only tangent, is it sufficient to say that f is concave all over his field? Justify.

MCQ A · diversion and convexity

After S29; detrievable function over each interval considered.

For each question, only one proposal is accurate. consult the correction.

1. If $f''(x) > 0$ over an interval, f is:

A. Decreasing B. Convex

C. Concave D. constant

2. The derivative of e^{1+x^2} is:

A. e^{1+x^2} B. $2xe^{x}$

C. $2xe^{1+x^2}$ D. $1+2x$

3. A tangent to a convex function is:

A. Always above B. still below the interval

C. parallel to the axis D. None

4. If $f'(x)$ change of sign of $+$ to $-$ in a, then f :

A. has a minimum in a B. has a local maximum in a

C. is convex D. is not continuous

5. The function $x \mapsto |x|$ is:

A. Derivable in 0 B. Not continuous in 0

C. continuous but not driftable in 0 D. Concave on $\mathbb{R}$

MCQ B · Continuity and VIT

After S33.

For each question, only one proposal is accurate. consult the correction.

6. To apply the VIT to $f(x) = 0$ on $[a; b]$, we check first:

A. that f is continuous on $[a; b]$ B. That f' is null

C. That $a = b$ D. That f is affine

7. If $f(a) < 0 < f(b)$ and f is continuous on $[a; b]$, then:

A. f has exactly one root B. f has at least one root

C. f is increasing D. $f(a) = f(b)$

8. To demonstrate the uniqueness of a solution, it is sufficient, for example, that f either:

A. Limited B. Strictly monotone

C. pair D. Derivable from a only

9. The dichotomy method maintains at each stage:

A. two bounds of the same sign B. an interval containing a root with signs opposing the bounds

C. a horizontal tangent D. an exact root value

10. A curve of f'' negative indicates that f' is:

A. increasing B. Decreasing

C. Positive D. None

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Any continuous function allows a root.

2. If f is convex, so f' is increasing where it is defined.

3. An approximate value obtained by dichotomy is evidence of existence.

Review review

I know how to associate sign of f'', convexity and variations in f' ; isolate and then justify a solution. **To be monitored:** announce uniqueness without monotonicity; confuse a graphic conjecture with the VIT.

Corrected MCQs

1. **B.** The sign $f'' > 0$ means that f' It does not give the sign of convexity. f' or the f.
2. **C.** The derivative of $e^{u(x)}$ is $u'(x)e^{u(x)}$. Here $u(x) = 1 + x^2$, therefore $u'(x) = 2x$. Forgetting this factor is a miscomposition.
3. **B.** A deviable convex function is above each of its tangents on the interval of It is not necessary for these tangents to be horizontal.
4. **B.** The positive sign of f' then negative gives growth and then decrease around a : maximum local, and not minimum.
5. **C.** $|x| \to 0$ when $x \to 0$, so continuity. The slopes to the left and right are worth -1 and 1, therefore not Differentiability in 0.
6. **A.** The VIT uses continuity throughout the interval. No null derivative is required; an interval is required. reduced does not give a change of sign.

7. **B.** Continuity and change of sign ensure existence. A function may have several roots: uniqueness requires an additional argument.
8. **B.** A strictly monotonous function is injective: a given value has at most one preimage. Being bounded or paired does not insure this property.
9. **B.** As long as no exact root is found, the sub-interval with the bounds is retained. images of opposite signs; its length is divided by two.
10. **B.** Over an interval where $f'' < 0$, the derivative of f' is negative, so f' His sign is still in study.

True/false. False: e^x Never cancels. True if f is derivable over the interval: it is a characterization
False: an approaching number alone does not prove existence; dichotomy must start
a time interval where continuity and change of sign are established.

Corrected applications and support

Application 1. g is defined on $[1/2;+\infty[$ and drawable on $]1/2;+\infty[$, $g'(x) = 1/\sqrt{2x-1} > 0$. h is derivable on $\mathbb{R}$, $h'(x) = -2xe^{1-x^2}$.

Application 2. The inner derivative is 3, so $f' = 12(3x+1)^3$. The coefficient x in the development $(1+3x)^4$ is 12, where $f'(0) = 12$, incompatible with the proposed formula.

Implementation 3. C is f, A is f' and B is f''. The zeros of A in ± 1 correspond to the extrema of C; the zero of B in 0 separates the decay and growth of A. Calculation: $f' = 3x^2 - 3$, $f'' = 6x$.

Implementation 4. The three are continuous. The polynomials are derivable; $|x|$ a side slopes -1 1. For x^4, $f'' = 12x^2 \geq 0$, therefore no inflection in 0; for x^3, $f'' = 6x$ change of sign, so 0 inflection. $|x|$ is convex on $\mathbb{R}$ and does not change convexity to 0.

Application 5. The staircase function never takes the value 0: it is not continuous in 0. x^2 is continuous but is not strictly monotonous on $\mathbb{R}$; $x^2 = 1$ a both solutions ± 1.

Implementation 6. Lay down $G(x) = e^{-x} - x$, continue, with $G(0) = 1 > 0$ and $G(1) = e^{-1} - 1 < 0$. $G' = -e^{-x} - 1 < 0$ ensures uniqueness. $G(1/2) > 0$, then $G(3/4) < 0$: interval $[1/2;3/4]$. Since $[0;1]$, seven steps give a width 1/128; six are not enough.

Guidance and review

Hide the formulas during the first work on the three curves. If the student confuses signs and change, point a portion of f negative but increasing. After a week, ask a new justification of uniqueness without the "use VIT" instruction.

Conduct and correction

Benchmarks for the teacher

The progression concentrates this chapter in three sessions: really divide S29 between diversions and convexity, then repeat the evidence of the tangents in S33. Continuous/derivable/convex vocabulary It should be distinguished by examples. Continuity gives existence by the VIT; strict monotonicity gives unity. For dichotomy, require an initial interval where the sign changes and a continuous function. A change of convexity, not just the cancellation of f'', characterizes a deflection point.

Corrected activity

1. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ and $f''(x) = 6x$. The sign of f' is positive on $]-\infty, -1[$ and $]1, +\infty[$, zero in -1, 1, negative between these values. f'' is negative to the left of 0, zero to 0, positive to Right.

2. Variations and limitations are summarized by

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

   The sign of f'' gives concavity on $]-\infty, 0]$ and convexity on $[0, +\infty[$.

3. Convexity changes to 0, and $f(0) = 0$: the point of inflection is $(0, 0)$. Its tangent is an equation $y = f(0) + f'(0)x = -3x$.

4. The function is continuous. $]-\infty, -1]$, it is strictly increasing $-\infty$ to 2: level 1 is reached exactly once. $[-1, 1]$, it strictly decreases from 2 to -2 : a second solution. $[1, +\infty[$, it grows strictly from -2 à $+\infty$: a third. Points -1, 1 are not no solutions, so no solution is counted twice.

5. Continuity provides the existence of a preimage for intermediate values. not unique: the function studied is continuous and level 1 has three preimage.

E1. For $u = 2x - 1$, $(u^5)' = 5u'u^4 = 10(2x-1)^4$. For $v = -x^2$, $(e^v)' = v'e^v = -2xe^{-x^2}$. Both functions are derivatable on $\mathbb{R}$.

E2. $f' = 4x^3 - 12x$ then $f'' = 12(x^2 - 1)$. The sign of f'' is positive outside of $[-1, 1]$negative inside. f is convex on $]-\infty, -1]$ and $[1, +\infty[$, concave on $[-1, 1]$. Changes in convexity give inflections $(-1, -5)$ and $(1, -5)$.

E3. The tangent is $y = f(1) + f'(1)(x-1) = e + e(x-1) = ex$. Since $f''(x) = e^x > 0$, the exponential is convex and its curve is above this tangent: $e^x \geq ex$ on $\mathbb{R}$.

E4. Poisons $g(x) = x^3 + 2x - 5$. This polynomial is continuous and $g'(x) = 3x^2 + 2 > 0$ on $\mathbb{R}$. In addition $g(1) = -2 < 0$ and $g(2) = 7 > 0$: the VIT gives a zero in $]1, 2[$, and strict growth on $\mathbb{R}$ insurance that it is unique.

E5. After n steps, length is $\frac{2}{2^n}$. The requested condition is equivalent to $2^n \geq 2000$. Like $2^{10} = 1024 < 2000 \leq 2048 = 2^{11}$, eleven steps are enough and ten steps are not enough.

Corrected from DM4

1. $f(0) = 1$ and $\lim_{x\to 0^+} f(x) = 1$. In $+\infty$, $f(x) = xe^{-x} + e^{-x} \to 0$ by comparative growth. product, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. This derivative is strictly negative for $x > 0$: f decrease strictly from 1 to 0 on $[0, +\infty[$.

2. $f''(x) = -e^{-x} + xe^{-x} = (x-1)e^{-x}$. So f is concave on $[0,1]$, convex $[1, +\infty[$. The change of convexity to 1 gives the point of inflection $(1, \frac{2}{e})$.

3. f is continuous and strictly decreasing on $[0, +\infty[$; $f(0) = 1 > \frac{1}{2}$ and its limit 0 is lower to $\frac{1}{2}$. The corollary of the VIT gives a single solution α. In addition, $f(1) = \frac{2}{e} \simeq 0{,}7358 > \frac{1}{2}$ and $f(2) = \frac{3}{e^2} \simeq 0{,}4060 < \frac{1}{2}$, therefore $1 < \alpha < 2$.

4.
```
From math import exp

def root_dm4():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+1)*exp(-m) > 0.5:
            a = m
        Else :
            b = m
    return a, b
```

   The decline keeps $f(a) \geq \frac{1}{2} \geq f(b)$; α remains in $[a,b]$. After eleven steps, we get $[1{,}677734375, 1{,}6787109375]$, length $\frac{1}{1024} < 10^{-3}$. Value $\alpha \simeq 1{,}67835$ is compatible with This framework.

5. The tangent in 0 is $y = f(0) + f'(0)x = 1$. Variations $f(x) \leq 1$, so the whole curve is below this tangent. This does not characterize concavity: it should be below of all the tangents, while f becomes convex after 1.

Corrected work in the manual or an anal

For $f(x) = 5x^2e^x$, the derivation of a product gives

$$f'(x) = 10xe^x + 5x^2e^x = 5x(x+2)e^x.$$

Since $e^x > 0$, f' is positive on $]-\infty;-2[$ and $]0;+\infty[$negative on $]-2;0[$, none in -2 and 0. function thus grows up to $f(-2) = 20e^{-2}$, decreases until $f(0) = 0$, then grows. Its limits are 0 in $-\infty$ (combined growth) and $+\infty$ in $+\infty$. A new diversion gives

$$f''(x) = 5(x^2+4x+2)e^x = 5\big((x+2)^2-2\big)e^x.$$

Poisons $a = -2-\sqrt{2}$ and $b = -2+\sqrt{2}$. The polynomial factor is positive outside of $[a;b]$negative in $]a;b[$. Thus f is convex on $]-\infty;a]$ and $[b;+\infty[$, concave on $[a;b]$. The sign changes to both roots: the points of inflection are $(a;5a^2e^a)$ and $(b;5b^2e^b)$.

The table of variations is summarized as follows:

x	$-\infty$		-2		0		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	0	$\nearrow$	$20e^{-2}$	$\searrow$	0	$\nearrow$	$+\infty$

For question 5, the window below makes visible the two changes of convexity and both extrema. Marked points correspond to calculated inflection points; the plot is used to control consistency of the study.

